

AUTOMATED INVESTOR POWERBI DASHBOARD

Submitted To Prof.

Daitri Tiwary

Data Visualization, Ethics and Privacy: MPBA G511



Submitted By: **Group 5**

Name	BITS ID
Jayasuriya P	2025H1540850P
Vijay J	2025H1540806P
A Atul	2025H1540842P
Ayushi Aggarwal	2025H1540820P
Avatansh Pandey	2025H1540809P

Birla Institute of Technology and Science, Pilani Campus, India

AUTOMATED INVESTOR POWERBI DASHBOARD

Agentic Financial Intelligence Monitor – PowerBI Dashboard

Problem Statement:

Financial firms equity research team faces information overload and delays in analysis due to the need to process thousands of pages from annual reports, ESG filings, and management discussions while also tracking fast-moving market data such as price volatility and volume spikes from Angel One. Balancing qualitative insights with high-frequency trading signals and long-term fundamental trends makes it difficult to connect short-term market movements with underlying business performance.

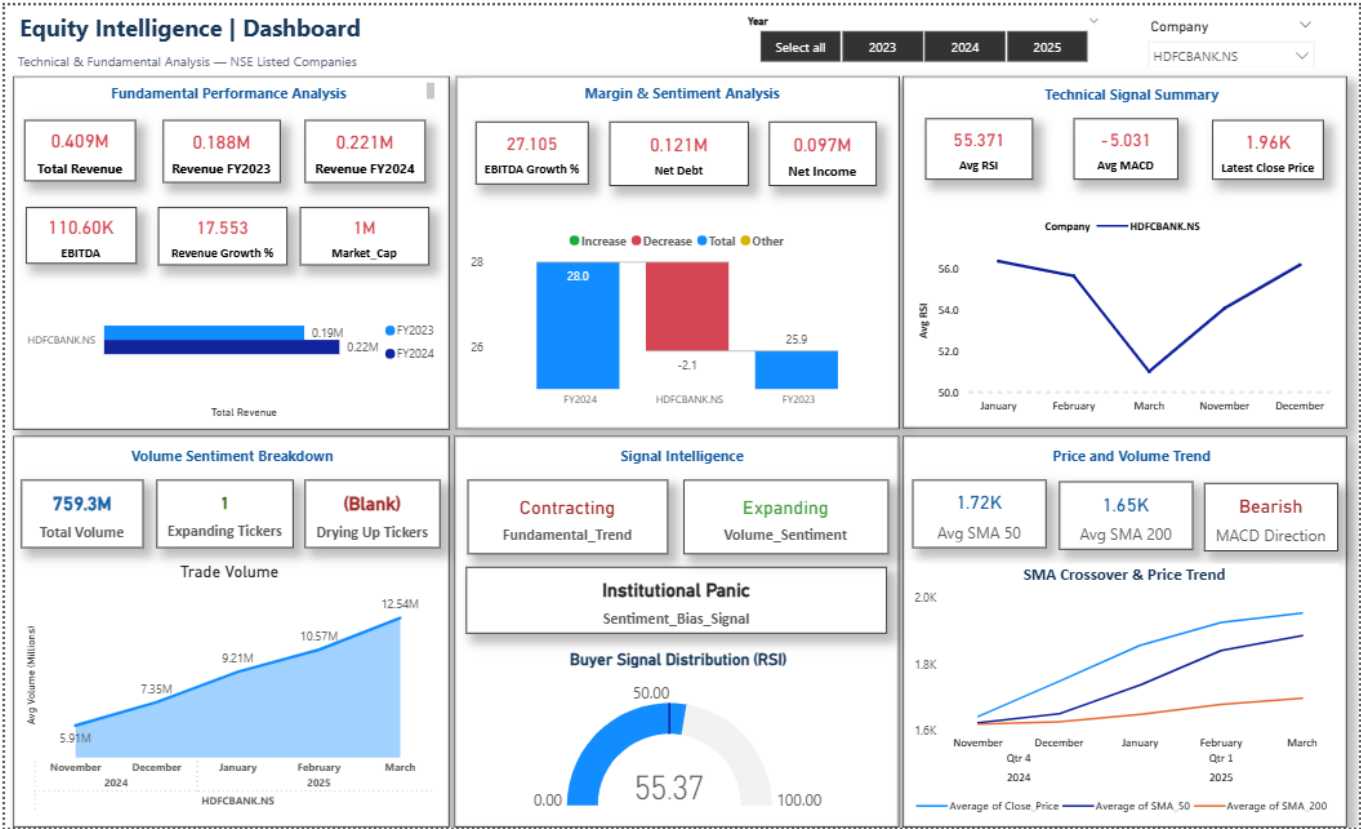
Project Summary:

- An Automating real-time market data ingestion via a free API
- Using a large language model (Gemini) to extract structured financial data from annual report PDFs
- Storing both data streams in a unified, relational CSV format
- Presenting synthesized signals in a single Power BI dashboard

Technology Stack:

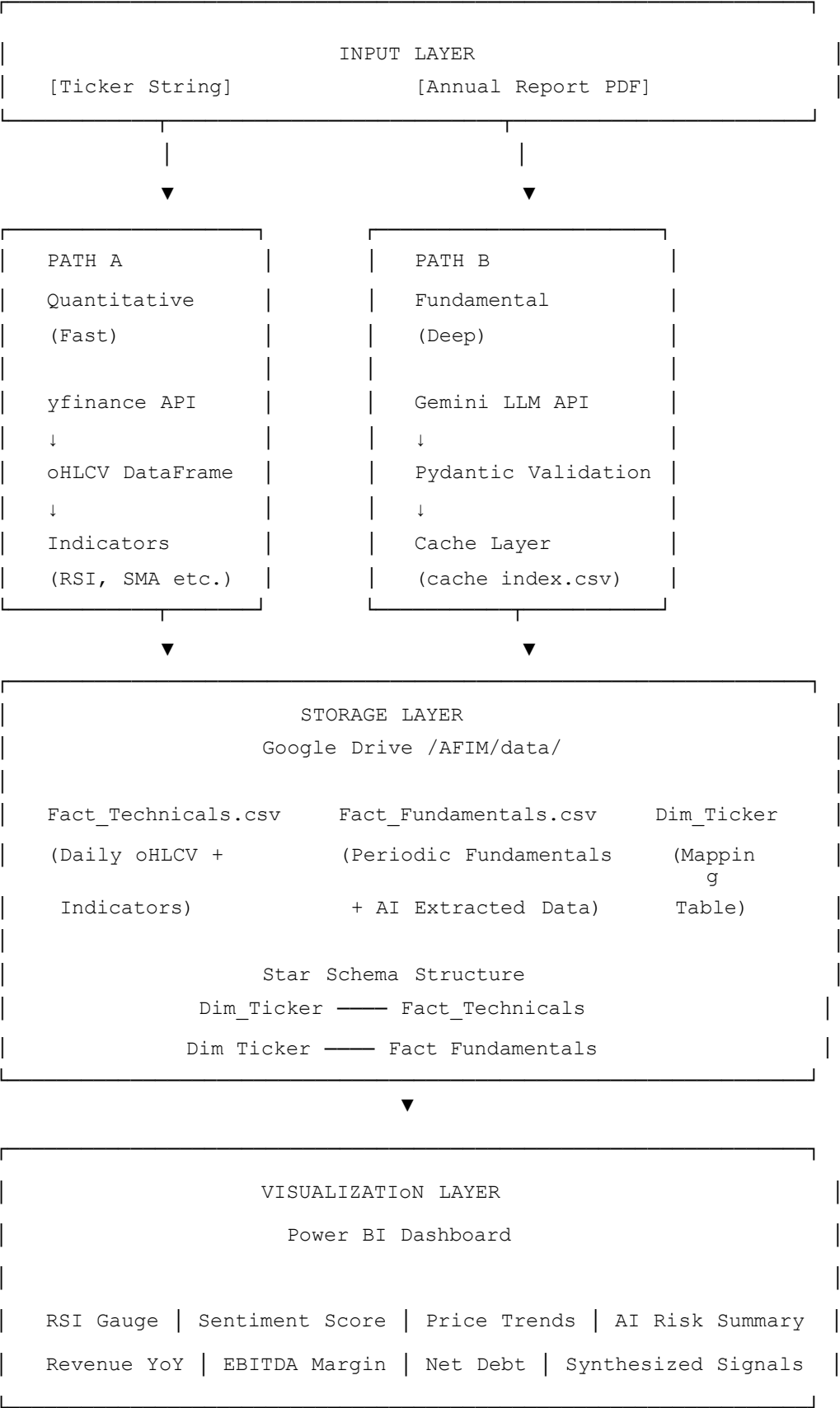
Component	Technology	Rationale
Runtime Environment	Google Colab	Cloud-based, iPad-accessible, free
Storage	Google Drive	Native Colab integration, Power BI connectable
Market Data	yfinance (Python)	Free, no API key, comprehensive OHLCV + metadata
LLM Engine	Google Gemini API (gemini-1.5-flash / gemini-2.5-flash)	Long context window, free tier available
Data Processing	Pydantic (Python)	Enforces schema on LLM output, prevents hallucination bleed
Technical Indicators	pandas-ta (Python)	Lightweight, no external dependencies
Caching	CSV-based (cache_index.csv)	Zero infrastructure, sufficient for single-user scale
Visualization	Microsoft Power BI	Star schema support, rich financial visuals, DAX engine
Development Language	Python	Universal data science standard

SNAPSHOT OF POWERBI DASHBOARD



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System Layer Architecture:

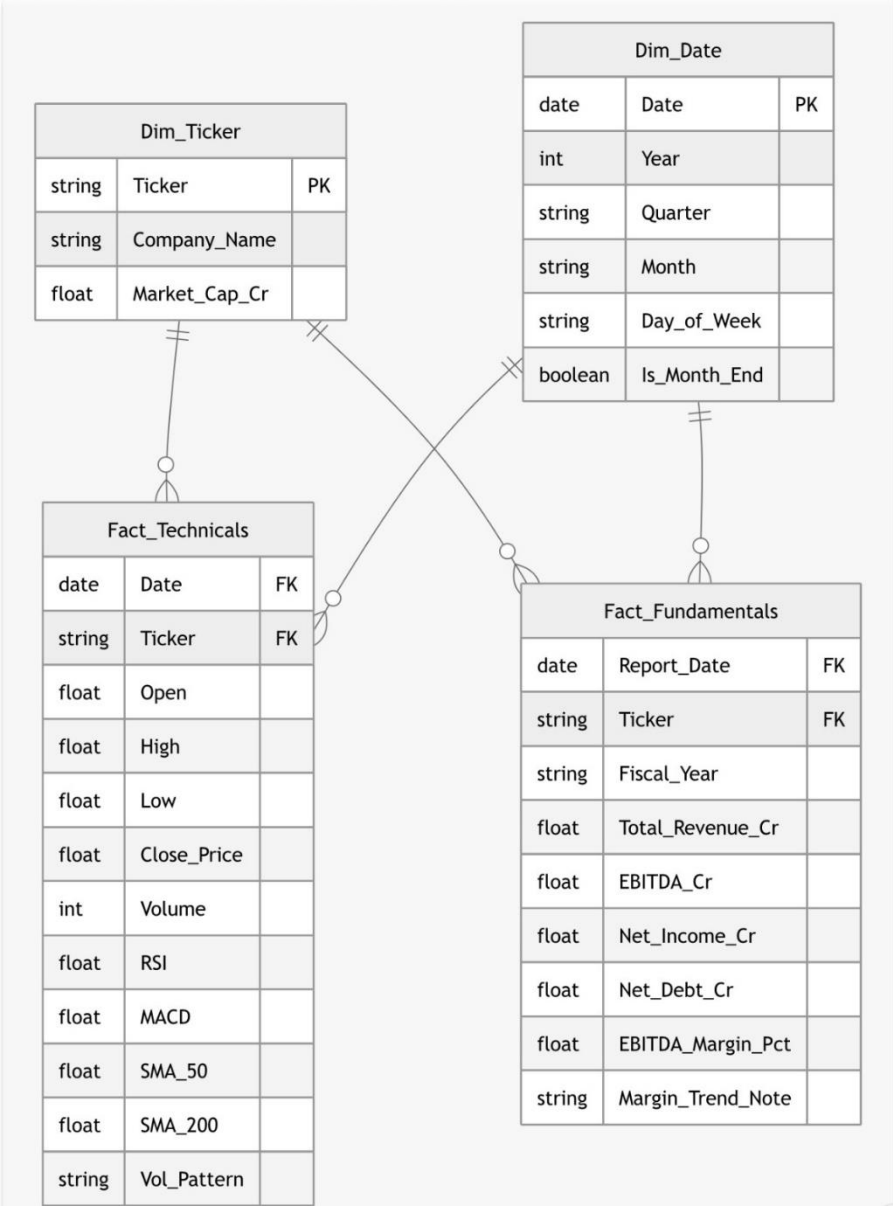
SYSTEM ARCHITECTURE CONTEXT

Path A — Quantitative yfinance API → RSI, MACD, SMA, Volume → Fact_Technicals.csv	Path B — Fundamental Gemini LLM → Pydantic validation → Fact_Fundamentals.csv	Synthesis Layer Power BI DAX Measures combine both paths into 3-signal output: Margin Trend + Volume Trend → Sentiment Bias Signal
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DATA ARCHITECTURE

Star Schema Design

Star schema — the standard for analytical/BI workloads — with one dimension table and two fact tables.



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Data Flow Description

- Path A** (Quantitative – Daily frequency) A ticker symbol is passed into the yfinance API, which returns historical and near-real-time OHLCV (Open, High, Low, Close, Volume) data. This data is transformed into a Pandas DataFrame, enriched with technical indicators, and written to Fact_Technicals.csv in Google Drive. This path runs daily and is the “fast” data stream.
- Path B** (Fundamental – Periodic/Annual frequency) An annual report PDF is uploaded and passed to the Gemini large language model API. Gemini’s long-context window (1M+ tokens) allows it to ingest an entire annual report in a single pass. A fixed prompt instructs Gemini to extract specific financial metrics. The response is validated against a Pydantic schema to enforce data structure and prevent hallucinations. The validated output is written to Fact_Fundamentals.csv. A cache layer prevents redundant API calls for previously processed documents.

ACTIONABLE INSIGHT SUMMARY

The dashboard reveals a structural divergence across the portfolio: **four of six tickers show improving EBITDA margins (Expanding), yet only three show rising trade volume.** This means the market has not yet priced in the fundamental improvement of TCS and INFY — creating a quantifiable mispricing opportunity. WIPRO is the inverse: volume is rising on deteriorating margins, signalling institutional exit rather than genuine confidence.

SIGNAL MATRIX — ALL 6 TICKERS

TICKER	M1: MARGIN	M2: VOLUME	M3: SIGNAL	IMPLICATION
RELIANCE.NS	Expanding	Expanding	Market Agreement	Strong Buy — both signals aligned
HDFCBANK.NS	Expanding	Expanding	Market Agreement	Strong Buy — highest SMA gap (4.2%)
TCS.NS	Expanding	Drying Up	Sentiment Trap	Hidden Gem — accumulate before breakout
INFY.NS	Expanding	Drying Up	Sentiment Trap	Hidden Gem — margin up, low attention
WIPRO.NS	Contracting	Expanding	Institutional Panic	Strong Sell — smart money exiting
AXISBANK.NS	Contracting	Drying Up	Dead Money	Avoid — no buyers, weak fundamentals

ACTIONABLE INSIGHT

WHAT	WHY	RECOMMENDED ACTION
TCS and INFY both show expanding EBITDA margins (FY23→FY24: +7% and +7.5% respectively) yet their volume ratios are 0.79× and 0.83× — recent trading activity is below the 50-day average. The market has not reacted to the fundamental improvement.	SAMEPERIODELASTYEAR confirms margin expansion. DATESINPERIOD rolling windows confirm volume contraction. The gap between improving business and falling market interest is the classic institutional accumulation setup before a price re-rating.	Accumulate TCS and INFY on price weakness. Set RSI < 50 as the entry trigger — current Avg RSI of 59 leaves room. Monitor Volume Sentiment: once Measure 2 flips to Expanding for these tickers, Measure 3 upgrades from Sentiment Trap to Market Agreement — confirming the re-entry signal.

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SUPPORTING EVIDENCE FROM DASHBOARD VISUALS

VISUAL	WHAT IT SHOWS	HOW IT SUPPORTS
Volume Sentiment bar chart	TCS avg vol 1.14M (recent) vs 1.46M (50-day)	Confirms Measure 2 = Drying Up — 0.79x ratio
Margin waterfall chart	FY23 → FY24 margin progression per ticker	TCS and INFY both show positive margin contribution
Signal Intelligence matrix	3-column M1/M2/M3 grid per ticker	Sentiment Trap cells in amber — ready to upgrade
RSI line chart	RSI 50–70 for TCS/INFY over Nov–Mar	Not overbought — entry opportunity exists below RSI 50
Price & Volume combo chart	Price rising on falling volume for TCS/INFY	Classic pre-breakout divergence pattern

CODE BUILD METHODOLOGY

Block Schedule:

Block	Name	Output
1	Market Data Engine	Fact_Technicals.csv + Dim_Ticker.csv live in Drive
2	Gemini Wrapper	PDF → validated JSON → Fact_Fundamentals.csv row
3	Cache Layer	Same PDF twice → Cache hit on second run
4	Schema Expansion	Real financial fields in both Fact tables
5	Unified Formatter	All CSVs Power BI-ready, star schema keys consistent
6	Power BI Connection	Dashboard loads all 3 tables, relationships confirmed